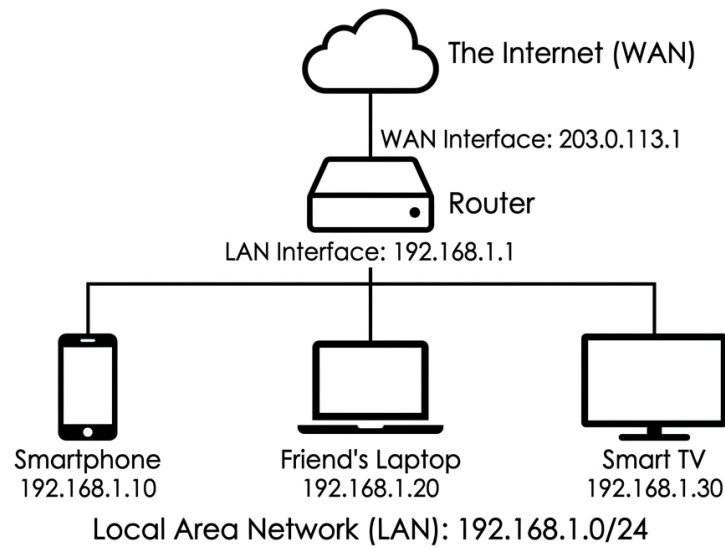


Network Diagram

Simple Home Network Diagram



Manual IP Routing Example

Laptop: IP Address: 192.168.1.10

Friend's Computer: IP Address: 192.168.2.20

Router 1: 192.168.1.1

Router 2: 192.168.2.1

Steps:

1. Laptop sends packet to 192.168.2.20.
2. Laptop checks routing table and finds destination is in another subnet.
3. Packet is sent to default gateway (192.168.1.1).
4. Router 1 checks its routing table.
5. Router 1 forwards packet to Router 2.
6. Router 2 finds destination network 192.168.2.0/24.
7. Router 2 delivers packet to friend's computer.

Sample Routing Table Check:

Laptop:

0.0.0.0/0 → 192.168.1.1

Router 1: 192.168.2.0/24 → Router 2

Router 2: 192.168.2.0/24 → Directly Connected

Routing Table for Zomato-Style Delivery Network

City Branch	Network Address	Subnet Mask	Next Hop	Interface
	-----	-----	-----	-----
Ahmedabad	10.1.0.0	255.255.255.0	Direct	Gig0/0
Surat	10.2.0.0	255.255.255.0	192.168.100.2	Gig0/1
Vadodara	10.3.0.0	255.255.255.0	192.168.100.3	Gig0/2

Comparison of Static and Dynamic Routing

Type	Pros	Cons
	-----	-----
Static Routing	Easy to configure, Low resource usage, More secure	Manual updates required, Not scalable, Difficult for large networks
Dynamic Routing (RIP/OSPF)	Automatic route updates, Scalable, Handles network changes automatically	More CPU usage, More memory usage, More complex configuration

Example:

A food delivery app like Zomato can benefit from OSPF because routes automatically update if a network link fails.

Sample OSPF Configuration

Router 1

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.255 area 0
```

Router 2

```
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
```

Explanation:

```
router ospf 1
```

Starts OSPF process number 1.

```
network 192.168.1.0 0.0.0.255 area 0
```

Adds network 192.168.1.0/24 to OSPF Area 0.

```
network 10.0.0.0 0.0.0.255 area 0
```

Advertises the link between routers.

```
network 192.168.2.0 0.0.0.255 area 0
```

Adds network 192.168.2.0/24 to OSPF Area 0.

Result:

Both routers learn routes automatically and can exchange network information.